

Heart Disease Analysis using Tableau

TEAM ID	SWTID-2026-8531
PROJECT TITLE	Heart Disease Analysis using Tableau
MAX MARKS	5 MARKS

CHAPTER – VII

FUNCTIONAL AND PERFORMANCE TESTING

7.1 – Performance Testing & Functional Testing

Step 1: Monitor Data Rendered to Database

- Open. MySQL Workbench
- Select the heart disease database.
- Expand the tables and choose the required table.
- Click the (i) Information icon.
- Check details such as **table rows, column count, and storage information.**
- Verify that only the required amount of data is being retrieved to improve performance.

Step 2: Optimize Data Retrieval

- Ensure database queries load only necessary records.
- Reduce unnecessary data processing to avoid system overload.
- Improve dashboard response time and efficiency.

Step 3: Apply Data Filters in Tableau

- Create filters to limit the data displayed.
- Use filters on **Gender, Age, Race, Diabetes, Stroke, Smoking Status, and General Health.**
- Enable users to analyse specific segments of heart disease data.

Step 4: Verify Filter Utilization

- Apply individual and multiple filters.
- Check whether the dashboard updates quickly after filter selection.
- Ensure filtered data is displayed accurately.
- Confirm that filters improve user interaction and dashboard responsiveness.

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Step 5: Review Calculated Fields

- Open the Tableau Data Pane and check for calculated fields.
- In this project, no calculated fields have been created.
- If required, create calculated fields such as:
 - BMI Category
 - Risk Classification
 - Health Status Indicators
- Verify that calculated fields do not impact dashboard performance.

Step 6: Test Visualizations

Validate the performance of the following charts:

- Gender-wise heart disease
- Age-wise heart disease
- People Suffering from Diabetes and Stroke
- Impact of Smoking and Alcohol Drinking on Heart Disease
- Other Diseases vs Stroke
- Race-wise heart disease
- General Health vs heart disease
- Physical Activity vs heart disease
- Age and BMI vs heart disease
- People with Stroke Suffering from Diabetes and Heart Disease

Step 7: Verify Dashboard Performance

- Check whether visualizations load quickly.
- Ensure interactive filters respond efficiently.
- Confirm that the dashboard provides accurate and meaningful insights.
- Monitor dashboard loading time and responsiveness during user interaction.